Lecture 1

What a state is

Objectives. *A state assigns expectations to observables. In a separable Hilbert space, positivity, normalisation and countable additivity on orthogonal events make this rule a density operator. This includes infinitely many populated levels, but excludes an exactly sharp position.*

- Prove that normal expectation rules are exactly density operators.
- Prove that no density operator assigns an exactly sharp position.
- Compute a thermal oscillator's level probabilities and mean energy.

The inner product $\langle \varphi, \psi \rangle$ is antilinear in its first argument and linear in its second. The space $\mathcal{H}$ is a Hilbert space, whereas H denotes a Hamiltonian. The expectation rule is ω; angular frequency is ν. We use E for an event projection and reserve P for momentum. Tensor-factor order will be stated when a composite system is introduced.

Facts of analysis used in this course. The standing facts S1–S4 are recalled from *QM II*, §0.4. We also accept S5 and S6 below without proof. These facts specify which analysis may be used in an otherwise complete proof.

S1 The Fourier transform is unitary on $L^2(\mathbb{R}^n)$ and has the Fourier inversion formula, with QM-II's normalisation. For general L^2 inputs, inversion is understood in the L^2 sense.

S2 The Fourier functions $e^{2\pi i k x/L}/\sqrt{L}$, $k \in \mathbb{Z}$, and the sine functions $\sqrt{2/L}\sin(n\pi x/L)$, $n \geq 1$, are complete orthonormal bases of $L^2(0, L)$, where $L > 0$.

S3 Products of functions from orthonormal bases of $L^2(\mathbb{R})$ form orthonormal bases of $L^2(\mathbb{R}^2)$ and $L^2(\mathbb{R}^3)$.

S4 Smooth functions of compact support are dense in L^2 of an interval and of $\mathbb{R}^n$. Schwartz functions are dense in $L^2(\mathbb{R}^n)$ as well.

S5 A compact self-adjoint operator on a separable Hilbert space has an orthonormal eigenbasis, including a basis of its kernel. Its nonzero eigenvalues have finite multiplicity and can accumulate only at zero.

S6 Every bounded positive operator B has a unique bounded positive square root $\sqrt{B}$, with $(\sqrt{B})^2 = B$.

1.1 A rule for expectations

In *QM I*, a preparation chose a unit vector ψ_k with probability p_k. Every bounded observable A then had mean $\sum_k p_k \langle \psi_k, A\psi_k \rangle$. The density matrix collected those means into $\mathrm{Tr}(\rho A)$ (Definition 3.3 and Theorem 3.4); different preparations could give the same matrix. *QM II* treated a particle on a line by unit vectors of $L^2(\mathbb{R})$ and closed with a question: what replaces the density matrix when the space has infinitely many dimensions?

Start with the means themselves. We work on a nonzero separable complex Hilbert space $\mathcal{H}$: it has a finite or countable orthonormal basis (ONB). Write $B(\mathcal{H})$ for its bounded

operators. A bounded observable is a self-adjoint member of this space. An event is an orthogonal projection E; its expectation is its probability. Even for an unbounded observable, such as position Q, its spectral projections $E_Q(I)$ belong to $B(\mathcal{H})$. Knowing their probabilities determines the distribution of its readings. Its mean may still be infinite or undefined.

It is convenient to assign a complex number to every bounded operator: write $A = C + \mathrm{i}D$ with $C = (A + A^\dagger)/2$, $D = (A - A^\dagger)/(2\mathrm{i})$ self-adjoint, and extend the expectation by complex linearity.

Definition 1.1 (An expectation rule)**.** An *algebraic state* is a complex-linear map $\omega : B(\mathcal{H}) \to \mathbb{C}$ such that $\omega(B) \geq 0$ for every positive operator B, and $\omega(\mathbb{1}) = 1$. Here $B \geq 0$ means B is bounded self-adjoint and $\langle \psi, B\psi \rangle \geq 0$ for every $\psi \in \mathcal{H}$.

The letter ω denotes an expectation rule, never a frequency; our angular frequency is ν. Positivity means that a nonnegative observable has a nonnegative mean. Linearity is a requirement on expectations, including those of noncommuting observables; it does not assert that they have simultaneous values. We will add one continuity requirement before calling such a rule a *state*.

Lemma 1.2 (Cauchy–Schwarz for an expectation rule)**.** *For any algebraic state and $A, B \in B(\mathcal{H})$,*

$$|\omega(B^\dagger A)|^2 \leq \omega(B^\dagger B)\omega(A^\dagger A), \qquad |\omega(A)| \leq \|A\|\,. \tag{1.1}$$

Proof. For self-adjoint C, the inequalities $-\|C\|\,\mathbb{1} \leq C \leq \|C\|\,\mathbb{1}$ show that $\omega(C)$ is real. Decomposing A into its self-adjoint real and imaginary parts gives $\omega(A^\dagger) = \overline{\omega(A)}$. Put $a = \omega(A^\dagger A)$, $c = \omega(B^\dagger A)$ and $d = \omega(B^\dagger B)$. Positivity for every $z \in \mathbb{C}$ gives

$$0 \leq \omega((A + zB)^\dagger (A + zB)) = a + 2\,\mathrm{Re}(\bar{z}c) + |z|^2 d.$$

If $d > 0$, choose $z = -c/d$. Then $0 \leq a - |c|^2/d$, so $|c|^2 \leq ad$, as claimed. If $d = 0$, choose $z = -tc$ with $t > 0$. The expression becomes $a - 2t|c|^2$, which cannot stay nonnegative for all t unless $c = 0$. Thus the inequality also holds when the denominator would vanish. Finally take $B = \mathbb{1}$ and use $0 \leq A^\dagger A \leq \|A\|^2\,\mathbb{1}$. ∎

Thus $0 \leq \omega(E) \leq 1$, and finite additivity of event probabilities already follows from linearity. Countable additivity will be a separate condition, not a consequence of this lemma.

Self-check 1.1

1. Why can position probabilities be specified on $B(\mathcal{H})$ even though Q is unbounded?
2. Does $\omega(A + B) = \omega(A) + \omega(B)$ require $[A, B] = 0$?
3. Why does positivity imply $\omega(A^\dagger) = \overline{\omega(A)}$?

1.2 Positive trace class

The finite-dimensional trace is a sum of diagonal entries. In infinite dimension a positive operator's sum may diverge: the identity has one on every diagonal entry. We first build the part of trace theory needed for states. The two standing facts used here are S5, the spectral theorem for compact self-adjoint operators, and S6, existence of a positive square root. They are stated in the standing-fact list at the start of this lecture.

Definition 1.3 (Positive trace class and density operators)**.** A positive bounded operator ρ is *positive trace class* if, for one ONB (e_k), $\sum_k \langle e_k, \rho e_k\rangle < \infty$. This sum is its *trace*, $\operatorname{Tr}\rho$; for a divergent positive sum write $\operatorname{Tr}\rho = +\infty$. A *density operator* is a positive trace-class operator with $\operatorname{Tr}\rho = 1$.

The words "for one ONB" are enough: the next proposition proves that the sum is independent of the basis. A positive operator of infinite trace is not trace class. The infinite value is basis independent too: a finite sum in another basis would, by the proposition, make every basis sum finite.

Proposition 1.4 (The trace and its spectral form)**.** *Every positive trace-class operator is compact. It has an orthonormal eigenbasis (φ_j) with eigenvalues $p_j \geq 0$, and*

$$\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|, \qquad \operatorname{Tr}\rho = \sum_j p_j = \sum_k \langle e_k, \rho e_k\rangle \tag{1.2}$$

for every ONB (e_k). The operator series converges in operator norm. For a density operator, the p_j sum to 1.

For a density operator, this decomposition supplies one possible preparation; it does not single out how the state was prepared. Lecture 2 studies that freedom. The proof first turns a small diagonal tail into an operator-norm approximation by finite-rank operators. Only after this compactness step may we use S5 to choose an eigenbasis. Parseval's identity then compares the trace sums in different bases. This order avoids assuming that a general bounded positive operator already has an eigenbasis.

Proof. First choose the basis in the definition, and let $E_n = \sum_{k\leq n} |e_k\rangle\!\langle e_k|$. Put $R = \sqrt{\rho}$ (S6). For a finite sum $v = \sum_{k>n} c_k e_k$, the triangle inequality followed by the scalar Cauchy–Schwarz inequality gives

$$\|Rv\| \leq \sum_{k>n} |c_k|\,\|Re_k\| \leq \|v\| \left(\sum_{k>n} \langle e_k, \rho e_k\rangle\right)^{1/2}.$$

By continuity the same bound holds throughout $(\mathbb{1} - E_n)\mathcal{H}$. Also $\|Rw\|^2 = \langle w, \rho w\rangle \leq \|\rho\|\,\|w\|^2$. Since $\rho = R^2$, these two estimates give

$$\|\rho(\mathbb{1} - E_n)\|^2 \leq \|\rho\| \sum_{k>n} \langle e_k, \rho e_k\rangle \longrightarrow 0. \tag{1.3}$$

Thus ρ is a norm limit of the finite-rank operators ρE_n and is compact by the definition in *Linear Algebra*, Lecture 14: a compact operator is a norm limit of finite-rank operators.

S5 supplies an eigenbasis, including a basis of the kernel. Its eigenvalues are nonnegative and tend to zero if there are infinitely many nonzero ones, so the displayed spectral series converges in norm. For any ONB (e_k), Parseval gives

$$\sum_k \langle e_k, \rho e_k\rangle = \sum_k \sum_j p_j |\left\langle \varphi_j, e_k\right\rangle|^2 = \sum_j p_j \sum_k |\left\langle \varphi_j, e_k\right\rangle|^2 = \sum_j p_j.$$

All terms are nonnegative. Interchanging these sums means taking the supremum of the same finite partial sums; it requires no measure theory. In the initial basis the sum is finite, so all the others have that same finite value. ∎

For bounded A, define the expectation pairing by

$$\operatorname{Tr}(\rho A) := \sum_j p_j \left\langle \varphi_j, A\varphi_j\right\rangle. \tag{1.4}$$

It is absolutely convergent, with $|\operatorname{Tr}(\rho A)| \le (\operatorname{Tr}\rho)\,\|A\|$. It really is a trace: in any ONB, expanding ρ gives

$$\sum_k \langle e_k, \rho A e_k\rangle = \sum_j p_j \sum_k \langle e_k, \varphi_j\rangle \langle \varphi_j, A e_k\rangle = \sum_j p_j \langle \varphi_j, A\varphi_j\rangle .$$

For each j, Cauchy–Schwarz and Parseval bound the sum of the absolute values of the inner terms by $\|\varphi_j\|\,\|A^\dagger\varphi_j\| \le \|A\|$. The double sum is therefore absolutely convergent and can be reordered. In particular the pairing does not depend on which eigenbasis was chosen. It is linear in A, positive when $A \ge 0$, and equals 1 at $A = \mathbb{1}$ for a density operator. Every density operator thus gives an algebraic state.

For later lectures, $\mathcal{T}_1(\mathcal{H})$ denotes the complex span of the positive trace-class operators. Extend the trace linearly. This is well-defined: the absolutely convergent diagonal sums above compute each finite linear combination in any basis. No general trace-ideal theorem is needed in this lecture. This is the usual trace class, also characterised by $\operatorname{Tr}|T| < \infty$, where $|T| = \sqrt{T^\dagger T}$ (· *cited*; Hall, §19.2, pp. 421–422); the characterisation is not needed here.

Self-check 1.2

1. In an infinite ONB, is $\rho e_n = 2^{-n} e_n$, $n \ge 1$, a density operator? What about $\rho e_n = e_n/n$?

1.3 Normal states are density operators

If (E_k) are mutually orthogonal event projections, their sum is an operator defined by its action: $\sum_k E_k\psi = \lim_{N\to\infty} \sum_{k\le N} E_k\psi$ for each ψ. Indeed the finite sums satisfy $\sum_{k\le N} \|E_k\psi\|^2 \le \|\psi\|^2$ by Pythagoras, so their vector tails tend to zero. The series converges in $\mathcal{H}$. Its limit acts as the identity on each range and as zero on their common orthogonal complement, hence is the projection onto their closed span. This is a *strong* sum. Convergence need not hold in operator norm: for basis truncations E_N, $E_N\psi \to \psi$ but $\|\mathbb{1} - E_N\| = 1$ in infinite dimension. The bound in Lemma 1.2 alone cannot pass to such a limit.

Definition 1.5 (Normality)**.** An algebraic state is *normal* if every countable orthogonal family of projections satisfies

$$\omega\Big(\sum_k E_k\Big) = \sum_k \omega(E_k). \tag{1.5}$$

In this course, *state* means a normal algebraic state.

Normality is countable additivity of probabilities for exclusive events. In particular, if an ONB exhausts the possible alternatives, their probabilities must sum to 1. It rules out assigning probability zero to every one of those alternatives while assigning probability one to their union. It is an assumption about the expectation rule, not about any particular preparation.

Example 1.6 (Recovering a matrix from an expectation rule)**.** In the orthonormal basis e_0, e_1 of $\mathbb{C}^2$, suppose a rule gives $\omega(|e_0\rangle\langle e_0|) = 3/4$, $\omega(|e_1\rangle\langle e_1|) = 1/4$ and $\omega(|e_1\rangle\langle e_0|) = 1/8$. Taking adjoints gives $\omega(|e_0\rangle\langle e_1|) = 1/8$ too. If a density matrix represents this rule, direct multiplication of the finite trace shows that its row j, column k entry is $\omega(|e_k\rangle\langle e_j|)$: the order of the two basis labels is reversed. Thus

$$\rho = \begin{pmatrix} 3/4 & 1/8 \\ 1/8 & 1/4 \end{pmatrix}, \qquad \operatorname{Tr}\big(\rho(|e_0\rangle\langle e_1| + |e_1\rangle\langle e_0|)\big) = \frac{1}{4}.$$

The last value also follows directly by adding the two given rank-one expectations. For $\psi_+ = (e_0 + e_1)/\sqrt{2}$, expanding its projection gives $\omega(|\psi_+\rangle\langle\psi_+|) = (3/4 + 1/4 + 1/8 + 1/8)/2 = 5/8$. The diagonal entries alone would not determine this probability: the off-diagonal expectations matter. The finite trace therefore determines a candidate matrix directly. In infinite dimension an additional bound is needed to show that the candidate coordinates define a bounded operator. The next proof supplies it.

We construct the density operator from rank-one tests, then recover all bounded expectations by a tail estimate. The steps and their inputs are shown in Figure 1.1.

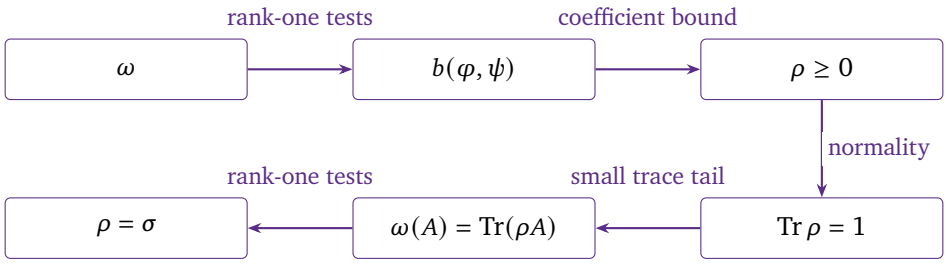


Figure 1.1: The construction from a normal expectation rule to its unique density operator. The form b collects rank-one expectations; normality makes the diagonal sum one, and the summable weights control every bounded A. Any density operator σ giving the same rule must equal ρ.

Theorem 1.7 (Normal-state representation). *On a nonzero separable complex Hilbert space, a functional ω is a state if and only if there is a unique density operator ρ such that*

$$\omega(A) = \mathrm{Tr}(\rho A) \qquad \text{for every } A \in B(\mathcal{H}).$$

Proof. A density operator gives a normal state. Only normality remains to be checked. Let $E = \sum_k E_k$ be a strong orthogonal sum. In the eigenbasis of ρ, orthogonality gives

$$\mathrm{Tr}(\rho E) = \sum_j p_j \left\|E\varphi_j\right\|^2 = \sum_j p_j \sum_k \left\|E_k\varphi_j\right\|^2 = \sum_k \mathrm{Tr}(\rho E_k).$$

The interchange again uses nonnegative terms.

Construct the operator from a state. The rank-one operators reveal all its matrix elements. Set

$$b(\varphi, \psi) = \omega(|\psi\rangle\langle\varphi|).$$

This form is antilinear in φ, linear in ψ, and $|b(\varphi, \psi)| \leq \|\varphi\|\,\|\psi\|$ by Lemma 1.2, since $\||\psi\rangle\langle\varphi|\| = \|\psi\|\,\|\varphi\|$. Fix an ONB (e_j) and a vector ψ. For $u_N = \sum_{j\leq N} b(e_j, \psi)e_j$ we have

$$\|u_N\|^2 = \sum_{j\leq N} |b(e_j, \psi)|^2 = b(u_N, \psi) \leq \|u_N\|\,\|\psi\|\,.$$

Consequently these squared coefficients sum to at most $\|\psi\|^2$. The series

$$\rho\psi = \sum_j b(e_j, \psi)e_j \tag{1.6}$$

converges and defines a linear bounded operator with $\|\rho\| \leq 1$. The coefficients in (1.6) are candidates for the coordinates of $\rho\psi$. The bound on their squared partial sums ensures that

they belong to the Hilbert space; merely assigning infinitely many matrix entries would not ensure this. Linearity follows because each coefficient $b(e_j, \psi)$ is linear in ψ. For a finite sum φ it satisfies $\langle \varphi, \rho\psi\rangle = b(\varphi, \psi)$; approximating an arbitrary φ by such sums proves the identity for all vectors. In particular, $\langle \psi, \rho\psi\rangle = \omega(|\psi\rangle\langle\psi|) \geq 0$. The identity $b(\varphi, \psi) = \overline{b(\psi, \varphi)}$, obtained by taking adjoints inside ω, shows that ρ is self-adjoint. Thus $\rho \geq 0$. This constructs ρ directly from an ONB, without an infinite-dimensional representation theorem for linear functionals.

Its trace is one. Normality on the rank-one basis projections gives

$$\sum_j \langle e_j, \rho e_j\rangle = \sum_j \omega(|e_j\rangle\langle e_j|) = \omega(\mathbb{1}) = 1.$$

By Proposition 1.4, ρ is a density operator.

Recover every bounded expectation. Now choose an eigenbasis (φ_j) of this ρ, including its kernel, and let $E_N = \sum_{j\leq N} |\varphi_j\rangle\langle\varphi_j|$. Since $|\varphi_j\rangle\langle\varphi_j| A = |\varphi_j\rangle\langle A^\dagger \varphi_j|$, the construction gives

$$\omega(E_N A) = \sum_{j\leq N} b(A^\dagger \varphi_j, \varphi_j) = \sum_{j\leq N} p_j \langle \varphi_j, A\varphi_j\rangle .$$

Cauchy–Schwarz for ω, with $B = \mathbb{1} - E_N$, gives the crucial tail bound

$$|\omega((\mathbb{1} - E_N)A)|^2 \leq \omega(\mathbb{1} - E_N)\omega(A^\dagger A) \leq \Big(1 - \sum_{j\leq N} p_j\Big) \|A\|^2 \longrightarrow 0. \tag{1.7}$$

Hence $\omega(A) = \lim_N \omega(E_N A) = \mathrm{Tr}(\rho A)$. For a finite-dimensional space the same argument terminates with $E_N = \mathbb{1}$.

Uniqueness. For any density operator σ, Equation (1.4) gives $\mathrm{Tr}(\sigma\,|\psi\rangle\langle\varphi|) = \langle\varphi, \sigma\psi\rangle$. If two density operators give the same rule, all their matrix elements agree, so they are equal. ∎

In $\mathbb{C}^n$, an orthogonal family has at most n nonzero projections, so linearity already implies normality. The theorem recovers *QM I*'s density matrices from rules for expectations, without first choosing an ensemble. A ray is included: $\rho = |\psi\rangle\langle\psi|$ for unit ψ gives $\omega(A) = \langle\psi, A\psi\rangle$. On an infinite-dimensional space the extra countable-additivity clause is essential. This is not peculiar to position space: every separable infinite-dimensional Hilbert space is unitarily equivalent to $L^2(\mathbb{R})$, by sending one ONB to another. The counterexample below therefore transfers to any such space.

For bounded A_N converging strongly to A, with a common bound $\|A_N\| \leq M$, the pairing also gives $\mathrm{Tr}(\rho A_N) \to \mathrm{Tr}(\rho A)$. Indeed split its eigenbasis series into a finite head and a tail: each head term converges, while the tail of the difference is bounded by $2M \sum_{j>J} p_j$ (the strong limit also has norm at most M). This is a bounded version of the strong continuity used as an axiom by Hall (Def. 19.6, p. 423). We derived it from normality; its finite-head and small-tail argument also handles position below.

Self-check 1.3

1. Where does normality first enter the construction of ρ?
2. Why does $E_N A \to A$ strongly not justify $\omega(E_N A) \to \omega(A)$ from the norm bound alone?
3. Verify $b(u_N, \psi) = \sum_{j\leq N} |b(e_j, \psi)|^2$. Which slot conjugates the coefficients?
4. Why would stopping after $\mathrm{Tr}\,\rho = 1$ leave the representation theorem unfinished? Explain the remaining finite-head and tail step.

1.4 An oscillator at temperature T

The harmonic oscillator of *QM II*, Lecture 8, has the complete Hermite basis $(h_n)_{n\geq 0}$ (Theorem 8.4) and energies $\hbar\nu(n+\frac{1}{2})$, with angular frequency $\nu > 0$. For $T > 0$, put $\beta = 1/(k_B T)$ and $q = e^{-\beta\hbar\nu}$, so $0 < q < 1$. The thermal weights are proportional to $e^{-\beta\hbar\nu(n+1/2)}$. Summing the geometric series gives

$$Z(\beta) = \sum_{n=0}^{\infty} e^{-\beta\hbar\nu(n+1/2)} = \frac{q^{1/2}}{1-q} = \frac{1}{2\sinh(\beta\hbar\nu/2)}, \qquad \rho_\beta = \sum_{n=0}^{\infty} (1-q)q^n \, |h_n\rangle\langle h_n| \,. \tag{1.8}$$

This is $e^{-\beta H}/Z(\beta)$, with the exponential defined on its eigenbasis. The series has nonnegative weights summing to 1, so it defines a density operator. It has infinitely many populated levels; finite rank is not required. The spectral event for energy $\hbar\nu(n+\frac{1}{2})$ is $|h_n\rangle\langle h_n|$, so the probability of level n is $(1-q)q^n$.

To sum its mean level number, let $S = \sum_{n\geq 0} nq^n$. The series is convergent for $q < 1$ (its successive nonzero terms have ratio tending to q). Subtracting qS from S yields $(1-q)S = \sum_{n\geq 1} q^n = q/(1-q)$. Therefore

$$\langle n\rangle = \frac{q}{1-q} = \frac{1}{e^{\beta\hbar\nu}-1}, \qquad \langle H\rangle = \hbar\nu\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\nu}-1}\right). \tag{1.9}$$

Here H is unbounded. Its mean is the mean of this energy-reading distribution. It is justified by the absolutely convergent series $\sum_n (1-q)q^n\hbar\nu(n+\frac{1}{2})$, not by inserting H into a theorem stated only for bounded A. As $T \downarrow 0$ the ground-state probability tends to 1 and the energy tends to $\hbar\nu/2$; for $k_B T \gg \hbar\nu$ the mean approaches $k_B T$ to leading order. The free particle on the whole line has no normalised Gibbs density operator (an example reserved for Sheet 1); the geometric-series argument depends on the oscillator's levels.

The physics

A trapped ion's motion in one approximately harmonic direction is an oscillator. Its motional temperature can be expressed as a mean level number. For example, a thermal preparation with $\nu/(2\pi) = 1\,\text{MHz}$ and $\langle n\rangle = 0.1$ corresponds to $T = \hbar\nu/[k_B \log(11)] \simeq 20\,\mu\text{K}$, and has ground-state probability $1/(1+\langle n\rangle) = 10/11$. These are illustrative model parameters, not a reported experiment. A mean level number alone does not establish that a preparation is thermal; the geometric level distribution is the additional model assumption.

Self-check 1.4

1. If the thermal mean level number is 1, what are q and the ground-state probability?

1.5 What $|x_0\rangle$ cannot be

On $L^2(\mathbb{R})$, *QM II* already ruled out a normalised position eigenvector (its closing system card, §15.4). Could a mixed state nevertheless assign an exactly sharp position? That would require $\text{Tr}(\rho f(Q)) = f(x_0)$ for every bounded continuous function f. The answer is still no.

Take continuous triangular bumps $f_n(x) = \max\{1 - n|x - x_0|/\ell, 0\}$, where $\ell > 0$ is a fixed length. They satisfy $0 \leq f_n \leq 1$ and $f_n(x_0) = 1$. The operator $f_n(Q)$ is multiplication by f_n. For any unit vector φ,

$$0 \leq \langle \varphi, f_n(Q)\varphi\rangle \leq \int_{|x-x_0|<\ell/n} |\varphi(x)|^2 \, \mathrm{d}x \longrightarrow 0.$$

Here is the elementary shrinking-interval step without an integration limit theorem: approximate φ in L^2 by a bounded continuous function v of compact support, using QM-II's density fact S4. On an interval I, the integral of $|\varphi|^2$ is at most $2\,\|\varphi - v\|^2 + 2|I|\,\|v\|_\infty^2$. First make the first term small, then shrink I.

For $\rho = \sum_j p_j \,|\varphi_j\rangle\langle\varphi_j|$, split the expectation into a finite head and tail. The head tends to zero by the preceding estimate; the tail is bounded by $\sum_{j>J} p_j$, uniformly in n. Hence

$$\operatorname{Tr}(\rho f_n(Q)) \longrightarrow 0, \qquad \text{whereas} \qquad f_n(x_0) = 1.$$

No density operator has the proposed evaluation rule. To state the existence result precisely, start on the algebra $\{f(Q) : f \in C_b(\mathbb{R})\}$ of bounded continuous multiplication operators. The assignment $f(Q) \mapsto f(x_0)$ is well-defined, positive, unital and has norm one, since $\|f(Q)\| = \sup_x |f(x)|$ for continuous f. It has a positive normalised extension to $B(L^2(\mathbb{R}))$ (the positive Hahn–Banach extension argument, · *cited*; Landsman, proof of Prop. C.15, p. 652, with Lemma C.4, p. 646). Every such extension is non-normal by the argument just given; it is outside this course's state convention. Thus the missing assumption is normality, even after we admit mixed states.

Self-check 1.5

1. Which property of a density operator makes the tail small? Is the exact value of its trace needed for this limit?
2. Could ever narrower wave packets contradict this result? Distinguish a fixed state from a changing preparation.

What is a state? First answer

A state is a positive, normalised linear rule for bounded expectations, countably additive on orthogonal events. On $B(\mathcal{H})$ for separable $\mathcal{H}$, it is exactly $\omega(A) = \operatorname{Tr}(\rho A)$ for one density operator. A sharp event $E_A(I)$ has probability $\operatorname{Tr}(\rho E_A(I))$. Unbounded means require an additional convergence statement. Neither a preparation ensemble nor a measurement update is specified by this definition.

Summary. Finite positive trace forces compactness, and the trace is basis independent. Positivity and Cauchy–Schwarz reconstruct an operator from rank-one expectations; normality makes its trace one and recovers all bounded expectations. The thermal oscillator is an infinite-rank example. An exactly sharp position cannot be represented by any density operator. Next we ask how many preparations one state can have.

Exercises. Try Sheet 1 in the separate *Exercise sheets* booklet (English).

Self-check answers.

1.1 1. The spectral projections of Q are bounded. 2. No; linearity concerns means, not simultaneous values. 3. Positivity makes the means of self-adjoint operators real; write $A = C + \mathrm{i}D$, so $A^\dagger = C - \mathrm{i}D$ with C, D self-adjoint.

1.2 The first has trace $\sum_{n\geq 1} 2^{-n} = 1$. The second is positive and bounded but has infinite trace; it is compact because its diagonal entries tend to zero. Compactness alone is insufficient.

1.3 1. In showing that the diagonal entries sum to 1. 2. Strong convergence does not imply norm convergence; the norm estimate cannot make the error small. Equation (1.7) supplies the missing control. 3. The first slot is antilinear, so each coefficient is multiplied by its complex conjugate. 4. The rank-one tests and trace do not yet identify every bounded expectation.

Split $A = E_N A + (\mathbb{1} - E_N)A$: the computed finite head has the trace value, and (1.7) makes the remaining expectation vanish.

1.4 $q = 1/2$ and probability $1/2$.

1.5 1. The convergent weight series has $\sum_{j>J} p_j \to 0$. Finite trace suffices for this limit; trace one normalises the probabilities. 2. No. The proof fixes one density operator before shrinking the bumps. Different narrow wave packets are different normal states; none has exactly sharp position. The limiting evaluation rule is non-normal.

Sources. K. Landsman, *Foundations of Quantum Theory* (2017), Def. 4.11 and Thm 4.12, pp. 109–110; the extension argument in Prop. C.15, p. 652, and Lemma C.4, p. 646; B. C. Hall, *Quantum Theory for Mathematicians* (2013), Defs. 19.6–19.7 and Thm 19.9, pp. 423–425. Hall uses sequential strong continuity in place of our projection-additivity axiom. The proof above constructs the representing operator by its ONB coefficients and uses Landsman's Cauchy–Schwarz tail estimate.